

AegisCare HMS — Role-Based User Manual

Document Title: Standard Operating Procedures & Role-Based User Manual

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Target Audience: Clinical, Administrative, and Financial Hospital Staff

Applicable Facilities: Kenya MoH Level 1–6 Facilities

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1. Receptionist

Task 1.1: Register a New Patient & Capture Digital Consent OTP

Who: `receptionist`, `records_officer`

Steps:

1. Log into AegisCare HMS and navigate to **Register Patient** (`/register-patient`) on the main navigation bar.
2. In the **Patient Identification** section, enter the patient's National ID, Passport Number, or Birth Certificate Number (for minors).
3. Fill in the demographic details:
 - Full Legal Name (First Name, Middle Name, Family Name).
 - Date of Birth (or check *Unknown DOB* and enter *Estimated Age*).
 - Gender / Biological Sex (**Male** / **Female**).
 - Mobile Phone Number (Must be formatted as **07...** or **01...** or **+254...**).
 - County of Residence and Residential Sub-County / Estate / Town.
 - Next of Kin Full Name, Relationship, and Contact Telephone.
4. In the **Consent Verification (ODPC Compliance)** section:
 - Click **Send Consent OTP SMS**.
 - An automated SMS containing a 6-digit numeric verification code is dispatched immediately via Africa's Talking to the patient's phone.
 - Request the 6-digit code from the patient and type it into the **Enter OTP Code** field.
 - Click **Verify & Accept Consent**. The status badge will change to a green **Verified & Consented** banner.
5. In the **Encounter & Payment Setup** section:
 - Select **Payment Mode**: **Cash**, **SHA SHIF / PHF**, **Private Insurance**, **Corporate Scheme**, or **Government Exemption**.
 - If Insurance or SHA: Select the provider and scheme, and input the Member / Policy Number.
 - Check the **Emergency Case** checkbox if the patient presents in acute clinical distress.
6. In the **Destination Room** dropdown, select the next operational station:
 - For Cash patients: Select **Triage** (or **Consultation Room** if pre-triaged).

- For Insurance / SHA patients: Select **Insurance Desk**.
 - For Acute Emergencies: Select **Resuscitation / Casualty Room**.
7. Click **Complete Registration & Open Encounter**.
 8. The system generates a permanent **Patient File Number** (e.g., **P000248**), opens an active encounter, generates a blank master invoice, and routes the patient's name to the live queue of the selected room.

Notes:

- ■■ *Emergency Override*: If a patient is unconscious, severely traumatized, or in shock and cannot receive an SMS OTP, check the **Emergency Intake Bypass** toggle. You must provide a brief mandatory justification note. The encounter will proceed immediately, and consent must be regularized upon patient stabilization.
 - Ensure the phone number is verified carefully; critical lab panic alerts and digital payment receipts are delivered to this number.
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Task 1.2: Check-In a Returning Patient (Revisit)

Who: **receptionist, records_officer**

Steps:

1. Navigate to **Patients** (**/patients**) or use the global top search bar.
2. Search for the returning patient by **File Number**, **National ID**, **Phone Number**, or **Name**.
3. Click on the patient record to open their master file.
4. Verify demographic details and update phone numbers or residential addresses if changed.
5. Click the **New Encounter** button in the top right corner.
6. Select the **Encounter Type** (**Outpatient, Specialist Clinic, Emergency**).
7. Confirm the **Payment Mode** and select the **Destination Room** (e.g., Triage, Insurance Desk, or MCH Clinic).
8. Click **Start Encounter**. The encounter is automatically categorized as an **OPD_REVISIT** for statutory MoH epidemiological reporting.

Notes:

- Never create a duplicate master file for an existing patient. Always search the central repository first to preserve longitudinal medical history.
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Task 1.3: Check-In an Appointment Booking

Who: **receptionist**

Steps:

1. Navigate to **Appointments** (**/appointments**).
2. Review the daily calendar or list view for scheduled patients.
3. Locate the arriving patient and verify their identity.
4. Click the **Check-In** action button next to their booking.
5. Confirm the payment mode and destination room.
6. Click **Confirm Arrival**. The system invokes **create_encounter_from_appointment**, transitions the appointment status to **checked_in**, creates an active clinical encounter, and sends the patient to the assigned clinician's waiting room.

Notes:

- If the patient fails to show up by clinic close, click **Mark No-Show** to release the clinician's calendar slot.
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2. Insurance Agent

Task 2.1: Process Insurance Clearance (SHA / Private / Corporate)

Who: **insurance_agent, accountant**

Steps:

1. Navigate to **Admin** → **Insurance Clearance** or open the **Insurance Desk** room queue (**/admin/insurance**).
2. Select a patient from the **Waiting for Clearance** queue.
3. Review the patient's insurance details:
 - Underwriter Name (e.g., Social Health Authority, Jubilee, Britam, CIC, APA).

- Scheme / Benefit Plan (e.g., SHA Outpatient, Corporate Comprehensive, Executive Inpatient).
- Policy / Member Number.

4. Eligibility Check:

- For SHA: Verify active member status on the SHA portal / NHIF legacy terminal.
- For Private / Corporate: Verify physical/digital smart card, biometric validity, and active member status on the insurer's portal.

5. Pre-Authorization (Preauth):

- If the visit involves specialized procedures, MRI/CT imaging, or inpatient admission, input the approved **Pre-Authorization Code** provided by the insurer into the preauth field.

6. Co-Payment:

- If the plan requires a fixed co-pay (e.g., KSh 500), verify the co-pay amount. The system automatically appends a co-pay line item to the patient's invoice.

7. Click Approve Clearance.**8. Select the destination room (e.g., Triage or Doctor Consultation Room) and click Dispatch Patient.**

- The encounter status changes to `insurance_status = 'cleared'`, unlocking doctor consultations, lab orders, and pharmacy dispenses under contracted insurance tariffs.

Notes:

- ■■■ *Visit Limit Warnings:* If a patient has exceeded their monthly or annual insurance visit cap, the system trigger `enforce_insurance_visit_limit` will display an on-screen alert. Do not approve clearance without documented authorization from the underwriter.

Task 2.2: Reject Clearance & Switch Patient to Cash Mode

Who: `insurance_agent`

Steps:

1. If the patient's insurance policy is lapsed, invalid, exhausted, or rejected by the underwriter:
2. Click **Reject Clearance** on the insurance clearance panel.
3. Select a structured rejection reason (e.g., `Policy Inactive / Expired`, `Benefit Limit Exhausted`, `Service Not Covered`, `Dependent Not Registered`).
4. Click **Convert Encounter to Cash**.
5. Inform the patient politely that the insurer has declined coverage.
6. Route the patient to the **Cashier / Triage** for cash-mode processing. The billing engine instantly switches price lookups from contracted insurance tariffs to standard cash prices.

Notes:

- Always document the underwriter's rejection reference code in the notes field for future dispute resolution.

Task 2.3: Track & View SHA / Insurance Claim Status by Date

Who: `insurance_agent`, `accountant`

Steps:

1. Navigate to **Insurance & SHA Claims** (`/admin/insurance`).
2. Use the date range picker to filter claims by admission, discharge, or submission date.
3. Filter claims by status:
 - `draft`: Encounter concluded, claim pending review.
 - `pending`: Queued in `dha_outbound_queue` awaiting transmission.
 - `submitted`: Transmitted to SHA / Insurer clearinghouse.
 - `approved`: Underwriter approved payment.
 - `rejected`: Underwriter query or rejection.
4. Click on any claim ID to inspect itemized diagnostic line items, attached ICD-11 codes, and the compiled FHIR Claim resource JSON bundle.

Notes:

- For queried claims, review the adjudicator notes, attach requested medical reports or lab results, and click **Re-Queue Claim**.

3. Triage Nurse

Task 3.1: Capture Patient Vital Signs & Acuity Assessment

Who: `triage_nurse, nurse`

Steps:

1. Open **Queue** (`/queue`) and switch to your assigned **Triage Room**.
2. Select the next patient listed under **Waiting** and click **Call Patient**.
3. Measure and enter physiological parameters:
 - **Blood Pressure (BP)**: Systolic and Diastolic in mmHg (e.g., `120 / 80`).
 - **Pulse / Heart Rate**: Beats per minute (e.g., `76 bpm`).
 - **Body Temperature**: Degrees Celsius (e.g., `36.8 °C`).
 - **Respiratory Rate**: Breaths per minute (e.g., `18 /min`).
 - **Oxygen Saturation (SpO2)**: Percentage on room air (e.g., `98 %`).
 - **Random Blood Sugar (RBS)**: In mmol/L (if clinically indicated or diabetic).
 - **Weight (kg) & Height (cm)**: System automatically calculates BMI and displays nutritional status.
4. **Pediatric Assessment (For Patients Under 5 Years)**:
 - Enter **MUAC (Mid-Upper Arm Circumference)** in centimeters:
 - Red (<11.5 cm): Severe Acute Malnutrition (SAM).
 - Yellow (11.5–12.5 cm): Moderate Acute Malnutrition (MAM).
 - Green (>12.5 cm): Normal.
5. **Clinical Acuity Tagging**:
 - Select **Triage Priority**: `Green (Routine)`, `Yellow (Urgent)`, `Red (Immediate Resuscitation)`.
6. Click **Save Vitals & Route Patient**.
7. Select the target **Consultation Room** (e.g., Doctor Room 1, Pediatric Clinic, MCH/FP Clinic, Dental) from the dropdown and click **Send to Doctor**.

Notes:

- **Panic Vitals**: If Systolic BP >180 mmHg, Temperature >39.5°C, or SpO2 <90%, the system flags a prominent red **Critical Alert**. Escort the patient immediately to the emergency resuscitation room and notify the casualty medical officer verbally.

4. Doctor / Clinical Officer

Task 4.1: Conduct Clinical Consultation & Review History

Who: `doctor, clinical_officer, dental_officer`

Steps:

1. Open **Queue** (`/queue`) and select your assigned **Consultation Room** (`/rooms/$id`).
2. The waiting list shows triaged patients prioritized by acuity (Red/Yellow/Green) and arrival time.
3. Select the patient and click **Start Consultation**.
4. Review the patient banner:
 - Demographics, Age, Gender, Payment Mode, Allergy Alerts.
 - Live Triage Vitals and computed BMI/MUAC.
 - Past medical history, chronic condition flags, previous clinic visits, and previous prescriptions.
5. In the **Clinical Notes** tab, document structured clinical findings:
 - **Chief Complaints (CC)**: Presenting symptoms and duration.
 - **History of Presenting Illness (HPI)**: Systematic chronological progression.
 - **Review of Systems (ROS) & Physical Examination (PE)**: General examination, systemic findings (CVS, RS, PA, CNS).
 - **Treatment Plan**: Intended clinical management strategy.
6. Click **Save Clinical Notes**.

Notes:

- ■ **Break-Glass Emergency:** If accessing a restricted or non-consented file during an emergency, click the **Break-Glass** shield button. Enter a mandatory medical justification (e.g., *"Unconscious road traffic accident victim requiring immediate neurotrauma review"*). The file will open instantly and record an immutable security audit event.

Task 4.2: Assign WHO ICD-11 Diagnosis Codes

Who: doctor, clinical_officer, dental_officer

Steps:

1. Scroll to the **Diagnoses** section on the consultation workbench.
2. In the **ICD-11 Search Bar**, begin typing the disease name, symptom, or code (minimum 2 characters, e.g., *"Malaria"*, *"Hypertension"*, *"Pneumonia"*).
3. The system queries WHO's live ICD-11 API via the **icd11-search** edge function and displays standardized international search results.
4. Click on the correct matching condition.
5. Select the **Diagnosis Type**:
 - **Primary:** The main condition responsible for this encounter.
 - **Secondary:** Co-existing medical conditions or chronic diseases.
 - **Working:** Provisional clinical hypothesis pending lab/radiology confirmation.
 - **Differential:** Alternative diagnostic possibilities.
6. Click **Add Diagnosis**. The condition is saved in **encounter_diagnoses** with its official ICD-11 MMS URI and description.

Notes:

- ■ **MOH Reporting Compliance:** Every encounter must possess at least one primary ICD-11 diagnosis before it can be legally signed.

Task 4.3: Order Diagnostic Investigations (Laboratory & Radiology)

Who: doctor, clinical_officer

Steps:

1. In the **Diagnostic Orders** tab:
 - **To Order Lab Tests:** Search the **lab_test_catalog** (e.g., *Full Hemogram*, *Malaria BS*, *Urinalysis*, *Lipid Profile*, *LFTs*). Check urgent priority if required. Add specific clinical instructions.
 - **To Order Imaging:** Select radiology modality and anatomical site (e.g., *Chest X-Ray PA*, *Abdominal Ultrasound*, *Head CT Plain*). Enter clinical indications.
2. Click **Submit Diagnostic Orders**.
3. The system instantly generates **lab_orders** and **radiology_orders**, attaches billable line items to the patient's invoice, and routes the patient's name to the respective Laboratory and Radiology queue screens.
4. Route the patient physically to the diagnostic departments.

Notes:

- When laboratory or imaging results are completed and verified, the patient will automatically re-appear in your room queue with a **Results Ready** notification badge.

Task 4.4: Prescribe Medications & Formulations

Who: doctor, clinical_officer, dental_officer

Steps:

1. Navigate to the **Prescriptions** section on the consultation screen.
2. Search for the required drug in the master formulary (e.g., *Amoxicillin/Clavulanic Acid 625mg Tab*, *Paracetamol 500mg Tab*, *Ceftriaxone 1g Inj*).
3. Specify the prescription parameters:
 - **Dose & Formulation:** e.g., **625 mg Tablet**.
 - **Frequency:** e.g., **TID (Three times daily)**, **BID (Twice daily)**, **OD (Once daily)**, **PRN (As needed)**.
 - **Duration:** Number of days (e.g., **7 days**).
 - **Route:** **Oral**, **Intravenous (IV)**, **Intramuscular (IM)**, **Topical**, **Inhalation**.
 - **Special Instructions:** e.g., *"Take with food"*, *"Complete full course"*.
4. Click **Add Prescription**.

5. Repeat for all required medications and click **Submit Prescriptions**.
6. The prescriptions are transmitted to the Pharmacy workbench, and invoice line items are generated automatically.

Notes:

- Review the patient's allergy alerts on the top header before prescribing penicillins, NSAIDs, or sulfa drugs.

Task 4.5: Admit Patient to an Inpatient Ward / ICU

Who: `doctor`, `clinical_officer`

Steps:

1. If the patient requires inpatient hospitalization, click the **Admit to Ward** button.
2. Select the target **Ward**:
 - *General Male Medical, General Female Surgical, Maternity, Pediatric, HDU, or Intensive Care Unit (ICU).*
3. Select an **Available Bed** from the live bed occupancy map.
4. Input the **Admission Diagnosis** (ICD-11) and author the initial **Admission Orders** (nursing instructions, IV fluids, monitoring frequency).
5. Click **Confirm Admission**.
6. The system executes the following:
 - Updates `admissions` table with status `admitted`.
 - Updates bed status to `occupied`.
 - If admitting to ICU: Automatically executes `charge_icu_admission_fee` to post the mandatory one-time ICU admission charge.
 - Routes the electronic chart to the Inpatient Nursing Station.

Notes:

- Ensure the ward selected is compatible with the patient's gender and age (e.g., adults cannot be admitted to Pediatric Ward).

Task 4.6: Finalize & Sign Clinical Encounter

Who: `doctor`, `clinical_officer`, `dental_officer` (Requires `sign_encounter` permission)

Steps:

1. Once all clinical documentation, diagnoses, lab reviews, and prescriptions are complete:
2. Click the green **Sign & Finalize Encounter** button at the bottom of the clinical screen.
3. The system checks validation constraints:
 - Confirms at least one valid ICD-11 diagnosis exists.
 - Confirms clinical notes are not empty.
4. A confirmation dialog appears: *"Are you sure you want to sign this encounter? This action will cryptographically lock all notes and diagnoses against further changes."*
5. Click **Sign Encounter**.
6. The system:
 - Sets `encounters.status = 'signed'`.
 - Records your digital signature, user ID, and exact timestamp.
 - Irreversibly locks clinical notes, diagnoses, and orders (`enforce_encounter_lock` trigger).
 - Triggers `auto_generate_sha_claim` for SHA-insured patients.
 - Queues FHIR R4 Encounter and Condition resources in `dha_outbound_queue` for national SHR synchronization.
 - Routes patient to **Pharmacy** (for drug collection) or **Cashier** (for final bill settlement).

Notes:

- **Immutability:** Once an encounter is signed, it cannot be edited by any staff member. Any supplementary clinical observations must be entered as an addendum in a new encounter.

5. Laboratory Technician

Task 5.1: Accession Specimens & Enter Laboratory Results

Who: `lab_tech`, `staff`

Steps:

1. Open the **Laboratory Workbench** (`/laboratory` or `/laboratory/$id`).
2. The queue displays pending diagnostic orders grouped by patient and urgency.
3. When the patient presents at the lab reception:
 - Verify patient identity against their printed lab requisition or file number.
 - Collect the designated biological sample (e.g., Venous Blood, Urine, Stool, Sputum, Swab).
 - Click **Collect Sample / In Progress**.
4. Perform the analytical assay on the designated laboratory machine or manual bench test.
5. In the laboratory results entry screen:
 - Input the quantitative values (e.g., Hemoglobin **13.4** g/dL, WBC **6.2** x10⁹/L, Platelets **250** x10⁹/L) or qualitative results (e.g., Malaria BS **No Malaria Parasites Seen**, Urinalysis Protein **Negative**).
 - The system displays standard reference ranges and highlights values outside normal limits with yellow or red flags.
6. Enter technologist remarks or specimen quality notes if applicable.
7. Click **Save Draft Results** to save progress, or proceed to verification.

Notes:

- For panic/critical results (e.g., Blood Glucose <2.2 mmol/L, Hemoglobin <5.0 g/dL, Potassium >6.5 mmol/L), call the ordering clinician immediately via internal phone in addition to logging the value in the system.

Task 5.2: Verify Results, Trigger Automated SMS & Auto-Return Patient

Who: `lab_tech`

Steps:

1. Review the entered test values for quality control and technical consistency.
2. Click **Verify & Publish Results**.
3. The system executes the following automated workflow atomically:
 - Updates `lab_results.status = 'verified'` and records `verified_by` and `verified_at`.
 - Locks results against further modification.
 - Invokes Edge Function `send-sms` via Africa's Talking to dispatch an automated notification SMS to the patient's phone: *"Dear Patient, your laboratory test results at [Hospital Name] are now ready. Please proceed to your consultation room."*
 - Executes `send_lab_results_to_requesting_room()`, automatically transferring the patient back into the requesting doctor's room queue.
 - Queues FHIR R4 **Observation** resources in `dha_outbound_queue` for DHA SHR transmission.

Notes:

- If a specimen is hemolyzed, clotted, or insufficient, click **Reject Specimen**, select the rejection reason, and the ordering clinician will be prompted to request a redraw.

6. Radiologist / Radiographer

Task 6.1: Process Radiology Order & Finalize Diagnostic Imaging Report

Who: `radiologist`, `doctor`

Steps:

1. Navigate to **Radiology** (`/radiology` or `/radiology/$id`).
2. Select a patient from the **Pending Imaging Queue**.
3. Position patient and perform requested modality scan (X-Ray, Ultrasound, CT, MRI, ECG, Mammography).
4. In the **Radiology Report Editor**:
 - Select the **Modality** and **Anatomical Site**.
 - Input structured findings:
 - *Clinical Indications:* (Pre-filled from doctor's order).
 - *Technique / Projection:* e.g., *"Standard Erect Postero-Anterior (PA) view of the chest"*.
 - *Detailed Observations:* Systematic organ review (e.g., lung fields, cardiac silhouette, bony thorax, diaphragmatic costophrenic angles).

- **Conclusion / Impression:** Concise diagnostic summary (e.g., "Right lower lobe consolidation consistent with lobar pneumonia. No pleural effusion.").

5. Click **Finalize & Sign Report**.

6. The system executes:

- Commits report to **radiology_results**.
- Sends automated completion SMS to the patient.
- Invokes **send_radiology_results_to_requesting_room()**, routing the patient back to the ordering doctor's consultation queue.

Notes:

- If additional contrast or specialized extra projections were required during the scan, add the supplementary consumables so they are billed accurately on the invoice.

7. Pharmacist

Task 7.1: Dispense Outpatient Prescriptions & Real-Time Stock Deduction

Who: **pharmacist, admin**

Steps:

1. Open the **Pharmacy Dispensing Workbench** (**/rooms/\$pharmacyRoomId**).
2. Select a patient from the **Pending Prescriptions** list.
3. Review the ordered medication list:
 - Drug Name, Strength, Dose, Frequency, Duration, and Special Instructions.
4. **Payment / Clearance Verification:**
 - Verify that the invoice line item for the prescription is marked **paid** (for Cash patients) or **cleared** (for SHA / Insurance patients).
 - If unpaid in cash mode: Direct the patient to the Cashier to settle the bill before dispensing.
5. In the dispensing panel, for each prescribed medication:
 - Select the active **Batch Number** and verify expiry date from the Central Pharmacy Store (**a00000000-0000-0000-0000-000000000003**).
 - Confirm available warehouse stock balance.
 - Pack the physical drugs, labeling each container with dosage, frequency, and cautionary instructions.
6. Click **Dispense Selected Medications**.
7. The system:
 - Updates **prescriptions.status = 'dispensed'** and stamps **dispensed_at** and **dispensed_by**.
 - Executes database trigger **dispense_prescription_stock()**, creating a negative movement in **stock_movements** and reducing physical stock balance.
 - Triggers **trg_medication_dispense_fhir** to compile a FHIR **MedicationDispense** resource for national health record sync.
8. Verbally counsel the patient on correct drug usage, potential side effects, and storage conditions. Hand over the medications.

Notes:

- **Stock Guard Violation:** If the pharmacy has zero available stock, the system will block dispensing. You must notify the storekeeper to execute an inter-store transfer from the Main Store or contact the clinician for an alternative therapeutic drug substitution.
- **ICU Segregation:** General Pharmacy cannot dispense ICU store medications (**sprint13b_dispense_stock_guard**). ICU ward nurses dispense ICU medications directly from the ICU sub-store.

8. ICU Nurse

Task 8.1: Maintain ICU Hourly Flow-Sheet & Critical Monitoring

Who: **nurse, icu_nurse**

Steps:

1. Open the **Inpatient ICU Module** or select the **ICU Clinical Room** (**a839ad3e-8721-4428-a825-6eee1f75207b**).

2. Select an admitted ICU patient bed.
3. Click **New Hourly Flow-Sheet Entry**.
4. Document the critical monitoring parameters:
 - **Hemodynamics:** Arterial Blood Pressure (Systolic, Diastolic, Mean Arterial Pressure - MAP), Central Venous Pressure (CVP), Heart Rate, Cardiac Rhythm.
 - **Mechanical Ventilation Settings:** Ventilation Mode (e.g., SIMV, PRVC, CPAP), Tidal Volume (VT), PEEP (cmH₂O), FiO₂ (%), Peak Inspiratory Pressure (PIP), Respiratory Rate.
 - **Neurological & Sedation:** Glasgow Coma Scale (GCS /15), Richmond Agitation-Sedation Scale (RASS: -5 to +4), Pupil size and light reactivity (Left/Right).
 - **Fluid Balance & Renal:** Hourly Urine Output (mL/hr), Gastric Aspirate / NG Drainage, Chest Tube Drainage, Central Line Infusions.
 - **Inotrope / Vasopressor Infusions:** Noradrenaline (mcg/min), Adrenaline, Dopamine, Dobutamine, Midazolam, Fentanyl infusion rates.
 - **Arterial Blood Gas (ABG):** pH, PaO₂, PaCO₂, HCO₃, Base Excess, Lactate, SaO₂.
5. Click **Save Hourly Record**. The chart compiles into a longitudinal real-time trend line.

Notes:

- Maintain strict hourly documentation. For MAP <65 mmHg, Urine Output <0.5 mL/kg/hr for 2 consecutive hours, or sudden desaturation, alert the Critical Care Physician immediately.

Task 8.2: Dispense Emergency Medications from ICU Sub-Store

Who: `icu_nurse`

Steps:

1. On the ICU Patient Chart, navigate to **ICU Bedside Dispensing**.
2. Select the required critical care drug from the dedicated **ICU Store** (`a99583cd-9354-470a-9299-73457734284d`).
3. Enter the quantity administered (e.g., *Noradrenaline 4mg Ampoule x 2*, *Propofol 1% 20mL Vial x 1*).
4. Enter clinical indication and administering nurse initials.
5. Click **Confirm ICU Administration**.
6. The system automatically logs stock consumption in `stock_usage`, updates the ICU stock ledger, and appends the medication line item to the patient's active inpatient invoice.

Notes:

- Emergency ICU medications are administered immediately without waiting for cashier payment clearance.

9. Dialysis Nurse

Task 9.1: Record Hemodialysis Session & Auto-Bill Consumables

Who: `dialysis_nurse`, `nurse`

Steps:

1. Open the **Dialysis Unit Room** (`/rooms/$dialysisRoomId`).
2. Select an arriving renal patient or click **Start New Dialysis Session**.
3. In the **Pre-Dialysis Assessment** section:
 - Record Pre-Dialysis Weight (kg), Dry Weight (kg), and computed Fluid Overload.
 - Record Pre-Dialysis Blood Pressure, Pulse, Temperature.
 - Assess Vascular Access: Type (**AV Fistula**, **AV Graft**, **Temporary IJ Catheter**, **Permacath**), Access Site, Thrill/Bruit status, and signs of infection.
4. In the **Machine & Prescription Parameters** section:
 - Assign Dialysis Machine Number (`machines` table).
 - Select Dialyzer Model / Surface Area (e.g., *High-Flux F60*, *Low-Flux F6*).
 - Set Blood Flow Rate (BFR in mL/min, e.g., **300**), Dialysate Flow Rate (DFR in mL/min, e.g., **500**).
 - Set Prescribed Treatment Duration (e.g., **4.0 hours**).
 - Set Ultrafiltration Goal (UF Goal in mL, e.g., **2500 mL**).
 - Specify Anticoagulation: Heparin Loading Dose (IU) and Maintenance Infusion (IU/hr) or Saline Flushes (for heparin-free dialysis).

5. In the Session Consumables Checklist:

- Check all single-use consumables utilized from the **Dialysis Store** (**d38d45c7-20f2-4e1b-9279-8cb5bf567cd1**):

- **Dialyzer Filter** (Qty: 1)
- **Arterial & Venous Bloodlines** (Qty: 1)
- **AV Fistula Needles 16G** (Qty: 2)
- **Acid Concentrate & Bicarbonate Cartridge** (Qty: 1)
- **Heparin 5000 IU/mL** (Qty: 1)
- **Normal Saline 1000mL** (Qty: 2)
- **Dialysis Dressing Pack & Gauze** (Qty: 1)

6. Monitor and document hourly intra-dialytic blood pressures, pulse, arterial pressure, venous pressure, TMP, and cumulative UF volume removed.

7. Upon session completion, document:

- Post-Dialysis Weight (kg), Net Ultrafiltration Removed (mL), Post-Dialysis BP.
- Any complications during session: Check applicable flags (*Hypotension, Cramps, Nausea, Access Clotting, Disequilibrium Syndrome*).

8. Click **Complete Session & Generate Billing**.

9. The system invokes **process_dialysis_session_billing**:

- Saves complete session clinical log in **dialysis_sessions**.
- Decrements all selected consumables from the Dialysis Store in **stock_usage**.
- Posts a comprehensive, itemized hemodialysis session bill to the invoice (or packages it for SHA Renal Replacement Claim submission).

Notes:

- For SHA-covered dialysis sessions, verify that the patient has not exceeded their 2 sessions/week statutory SHIF benefit allotment.

10. Accountant / Cashier

Task 10.1: Collect Payment, Issue Receipt & Allocate Funds

Who: **accountant, admin**

Steps:

1. Open **Billing & Accounting** (**/accounting** or **/invoices/\$id**).
2. Search for the patient by **Invoice Number**, **Patient File Number**, **National ID**, or **Name**.
3. Click on the active invoice to inspect all itemized line items (Consultations, Lab, Imaging, Pharmacy, Procedures, Bed Days).
4. Review the computed totals: **Total Amount**, **Waivers Applied**, **Amount Already Paid**, and **Net Outstanding Balance**.
5. Click **Record Payment**.
6. In the payment modal:
 - Enter **Amount to Pay** (KSh).
 - Select **Payment Method**: **Cash**, **M-Pesa**, **Credit / Debit Card**, **Bank Transfer**, **Insurance Guarantee / Letter of Undertaking**.
 - If M-Pesa / Card / Bank: Enter the unique **Transaction Reference Number** (e.g., M-Pesa Transaction ID **SDF8923JKL**).
7. Click **Confirm Payment**.
8. The system executes database trigger **recalc_invoice_payments()**:
 - Inserts immutable transaction record into **invoice_payments**.
 - Updates invoice **amount_paid** and recalibrates balance.
 - If balance is zero: Updates invoice status to **paid** and clears encounter locks.
 - Generates printable official hospital receipt with QR verification code.
 - Dispatches automated SMS payment confirmation to patient's mobile phone.

Notes:

- Never accept cash without generating and issuing an official printed system receipt immediately.

Task 10.2: Apply an Indigent Patient Waiver or Credit Note

Who: **accountant, admin** (Requires supervisory authorization)

Steps:

1. Open the target patient invoice in `/invoices/$id`.
2. Click the **Apply Waiver / Adjustment** button.
3. Enter the **Waiver Amount** (KSh).
4. Select the **Waiver Category**: **Indigent Patient**, **Hospital Board Waiver**, **Clinical Service Waiver**, **Billing Correction**.
5. Enter the mandatory **Supervisory Approval Reference** and justification text.
6. Click **Authorize Waiver**.
7. The system updates `invoices.waiver_amount`, logs `waiver_approved_by` with your user ID and timestamp, recalculates the net balance, and writes an audit event to `audit_log`.

Notes:

- **Audit Scrutiny:** All waivers are audited during institutional financial reviews and MoH accounting inspections. Never apply a waiver without written executive management approval.
-

Task 10.3: Execute End-of-Day (EOD) Cash Reconciliation

Who: `accountant, director`

Steps:

1. At the close of each cashier shift (e.g., 17:00 or shift change), navigate to **Accounting** → **End-of-Day Reconciliation**.
 2. Select your cashier user name and today's date range.
 3. The system compiles total collections categorized by channel:
 - Total Cash Collected (KSh)
 - Total M-Pesa Receipts (KSh)
 - Total Credit Card / Electronic Funds (KSh)
 - Total Waivers Applied (KSh)
 - Total Outstanding Invoices (KSh)
 4. Count physical currency in the cash drawer and enter the **Physical Cash Count** amount.
 5. The system calculates any variance between physical cash and system receipts.
 6. Click **Generate & Close EOD Shift Report**.
 7. Print the reconciled EOD summary, attach banking deposit slips, sign the report, and submit to the Chief Accountant / Hospital Director.
-

11. Mortician

Task 11.1: Admit an Internal Hospital Body to Mortuary

Who: `mortician`

Steps:

1. Open the **Mortuary Department** room (`214aca6c-cbde-40ce-af5a-92cc7815bda9`).
 2. Click **Admit Deceased Body**.
 3. Select **Body Source**: **Internal Hospital Death**.
 4. Search for the deceased patient by File Number or Inpatient Admission ID.
 5. Demographic details, date/time of death, ward of death, and certifying medical officer are auto-populated.
 6. Assign **Mortuary Cold Storage Unit / Vault Number** (e.g., **Cold Room B - Slot 14**).
 7. Enter receiving mortician notes (presence of valuables, personal effects, physical body condition).
 8. Click **Admit to Mortuary**.
 9. The system executes `assign_mortuary_reference()` (generating a unique mortuary reference, e.g., **MORT-2026-0042**) and links the mortuary stay to the patient's existing hospital invoice.
-

Task 11.2: Admit an External Body (Police Case / Home Death)

Who: mortician

Steps:

1. In the Mortuary screen, click **Admit Deceased Body**.
2. Select **Body Source:** External (Brought in Dead / Police Case / Home Death).
3. Input deceased demographic details: Full Name (or *Unknown Deceased Male/Female*), Estimated Age, Gender.
4. Input Intake Legal Metadata:
 - Brought In By: Name, ID Number, Phone, Relationship / Title.
 - If Police Case: Accompanying Police Officer Name, Rank, Service Number, Police Station, and Police OB (Occurrence Book) Number.
 - Place of Death, Date and Time of Death.
5. Assign **Cold Storage Unit / Slot**.
6. Click **Register External Body**.
7. The system:
 - Creates a record in `mortuary_records`.
 - Generates mortuary reference number.
 - Invokes `create_external_mortuary_invoice()` to provision a dedicated commercial mortuary invoice linked to the next-of-kin / police file.

Task 11.3: Document Mortuary Services & Release Body

Who: mortician, accountant

Steps:

1. As mortuary services are rendered during storage:
 - Click **Add Mortuary Service** on the deceased record.
 - Add billable services: *Embalming / Preservation, Post-Mortem Examination Attendance, Refrigeration Storage Days, Body Dressing & Beautification, Hearse Transfer*.
 - The system automatically appends line items to the mortuary invoice.
 - Nightly daemon `accrue_daily_mortuary_charges()` automatically accrues standard daily refrigeration storage fees.
2. **Body Release Procedure:**
 - Next-of-kin presents official **Burial Permit / Death Certificate** and national ID.
 - Next-of-kin proceeds to Cashier to settle the final mortuary invoice balance.
 - Mortician verifies invoice status is **paid** (or authorized waiver attached).
 - Click **Release Body**.
 - Input collecting next-of-kin name, national ID, vehicle registration, and burial destination.
 - Click **Confirm Body Handover & Discharge**. The storage slot is marked **available** and the mortuary record is officially closed.

Notes:

- **Release Lock:** The system will strictly block body release if an outstanding financial balance exists without an authorized waiver.

12. Store Keeper

Task 12.1: Receive Commercial Supplier Delivery into Main Store

Who: store_keeper, admin

Steps:

1. Navigate to **Deliveries & Stock Intake (/deliveries)**.
2. Click **New Delivery / Stock Receipt**.
3. Enter Delivery Metadata:
 - Supplier / Vendor Name (e.g., *KEMSA, MEDS, Harleys Ltd*).
 - Supplier Invoice Number / Delivery Note Number.
 - Local Purchase Order (LPO) Reference Number.
 - Receiving Location: Select **Main Store** (`a0000000-0000-0000-0000-000000000001`).
4. In the **Delivered Items** table, for each delivered commodity:
 - Search `stock_items` by name or SKU.

- Enter **Quantity Received** (Pack count / Units).
- Enter **Batch / Lot Number** and **Expiry Date**.
- Enter Unit Buying Price (KSh) and confirm selling tariff.
- 5. Click **Post Delivery to Stock**.
- 6. The system executes `delivery_to_stock()`:
 - Records delivery in `stock_deliveries`.
 - Creates positive ledger entries in `stock_movements`.
 - Updates stock item current available quantities and logs weighted average buying costs.

Notes:

- Carefully check physical seals, batch numbers, and expiry dates against the delivery note before clicking submit. Never accept stock expiring within 6 months without supervisor sign-off.
-

Task 12.2: Transfer Stock to Departmental Sub-Stores

Who: `store_keeper`, `pharmacist`

Steps:

1. Navigate to **Stock Management** (`/stock`).
 2. Click **Inter-Store Stock Transfer**.
 3. Select **Source Location**: `Main Store`.
 4. Select **Destination Location**:
 - `Central Pharmacy Store` (`a0000000-0000-0000-0000-000000000003`)
 - `ICU Store` (`a99583cd-9354-470a-9299-73457734284d`)
 - `Dialysis Store` (`d38d45c7-20f2-4e1b-9279-8cb5bf567cd1`)
 - `Laboratory Store`
 5. Select the commodity and input the **Transfer Quantity**.
 6. Add transfer note (e.g., "Weekly stock requisition for ICU").
 7. Click **Execute Stock Transfer**.
 8. The system invokes `transfer_stock_between_locations()` RPC:
 - Deducts quantity from Source Store in `stock_movements`.
 - Credits quantity to Destination Store in `stock_movements`.
 - Generates printable inter-store transfer voucher.
-

13. System Administrator (Admin)

Task 13.1: Manage Staff User Accounts, Roles & Permissions

Who: `admin`, `system_admin`

Steps:

1. Navigate to **Admin** → **Users & Access** (`/admin/users`).
2. To approve a newly registered staff member:
 - Locate user under **Pending Account Approvals**.
 - Click **Approve Account** (`is_approved = true`).
3. To assign roles:
 - Click **Edit Roles** on the user card.
 - Assign applicable roles: `doctor`, `clinical_officer`, `nurse`, `triage_nurse`, `lab_tech`, `radiologist`, `pharmacist`, `accountant`, `insurance_agent`, `mortician`, `store_keeper`, `records_officer`, `admin`.
 - Click **Save Role Assignments**.
4. To grant room access:
 - Navigate to **Admin** → **Rooms** (`/admin/rooms`).
 - Select consultation room or diagnostic unit and assign user to `user_room_access`.
5. To configure practitioner council details:

- In user profile, enter **Professional Council Type** (**KMPDC**, **COC**, **NCK**, **PPB**, **KMLTTB**) and **Council Registration Number** for statutory encounter signing.

Notes:

- Follow the principle of least privilege. Never assign **admin** or **sign_encounter** permissions to unauthorized non-clinical personnel.

Task 13.2: Configure Facility Metadata, Wards, Beds & Tariffs

Who: **admin**

Steps:

- 1. Facility Settings (/admin/settings):**
 - Verify Facility Name, KMHFL Code, SHA Facility ID, MoH Level (1–6), County, Contact Phone, and Official Email. Click **Save Settings**.
- 2. Wards & Beds (/admin/wards):**
 - Create inpatient wards, specify ward category, and assign standard **Daily Bed Rate (KSh)**.
 - Add individual physical bed numbers to each ward.
- 3. Master Pricing Catalog (/admin/services & /admin/pricing):**
 - Set standard cash prices for general consultations, specialty clinics, procedures, and nursing services.
 - Configure **Insurer Contracted Tariffs (insurance_contracted_prices)** to establish contracted rate overrides for SHA, Jubilee, Britam, and other underwriters.

14. Medical Director / Executive Management

Task 14.1: Review Executive Dashboard, Epidemiological Trends & Health

Who: **director**, **admin**

Steps:

- 1.** Log into the system and navigate to **Dashboard (/dashboard)**.
- 2. Clinical KPI Cards:**
 - Review Today's Outpatient Attendance, Total Active Inpatient Admissions, Bed Occupancy Rate (%), Emergency Casualty Volume, and Lab/Radiology Turnaround Times.
- 3. MOH Epidemiological Surveillance:**
 - Inspect the **Top 10 Morbidity Disease Chart** generated by `dashboard_top_diseases()` RPC to detect community disease outbreaks (e.g., spikes in Malaria, Gastroenteritis, or Upper Respiratory Infections).
 - Review the 7-day OPD vs. Inpatient Admission Trend graph (`dashboard_admitted_opd_trend()`).
- 4. Financial Summary:**
 - Inspect daily gross revenue, cash collections, insurance claims submitted, outstanding aging balances, and total waivers approved.
- 5. System Health & Integrations:**
 - Check status badges for Supabase Database connectivity, Africa's Talking SMS credit balance, and DHA Outbound Queue transmission latency.

Task 14.2: Generate & Export Statutory MoH Reports (705A, 705B, 717)

Who: **director**, **records_officer**, **admin**

Steps:

- 1.** Navigate to **MOH Reporting Suite (/moh or /moh/705, /moh/717)**.
- 2.** Select the target **Reporting Period** (Month and Year, e.g., *July 2026*).
- 3.** Select Report Type:
 - **MOH 705A:** Under 5 Years Outpatient Morbidity Summary.
 - **MOH 705B:** Over 5 Years Outpatient Morbidity Summary.
 - **MOH 706:** Laboratory Diagnostic Return.
 - **MOH 707:** Inpatient Bed Utilization & Mortality Return.
 - **MOH 717:** Institutional Workload Return.

4. Click **Compile MoH Report**.
5. The system executes `get_moh_705_report(start_date, end_date, form_type)` against database aggregate tags.
6. Review the tabular return displaying disease classification, ICD-11 codes, male cases, female cases, and total incidence.
7. Click **Export to CSV / Excel** or **Print Official MoH PDF** for submission to the Sub-County Health Records Information Officer (SCHRIO) and upload to the national DHIS2 / KHIS portal.